AL-2001
Artificial Intelligence
Lab # 2

Objectives:
- Strings, Functions

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. Do not copy from any source otherwise you will be penalized with negative marks.
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

Problems:

1. Write a program to iterate the first 10 numbers, and in each iteration, print the sum of the current and previous number.

Expected Output:

```
Printing current and previous number sum in a range(10)

Current Number 0 Previous Number  0  Sum:  0

Current Number 1 Previous Number  0  Sum:  1

Current Number 2 Previous Number  1  Sum:  3

Current Number 3 Previous Number  2  Sum:  5

Current Number 4 Previous Number  3  Sum:  7

Current Number 5 Previous Number  4  Sum:  9
```

```
Current Number 6 Previous Number  5  Sum:  11

Current Number 7 Previous Number  6  Sum:  13

Current Number 8 Previous Number  7  Sum:  15

Current Number 9 Previous Number  8  Sum:  17
```

2. Write a program to accept a string from the user and display characters that are present at an even index number.

For example, `str = "pynative"` so you should display ‘p’, ‘n’, ‘t’, ‘v’.

3. Write a program to remove characters from a string starting from zero up to `n` and return a new string.

For example:

- `remove_chars("pynative", 4)` so output must be `tive`. Here, we need to remove the first four characters from a string.
- `remove_chars("pynative", 2)` so output must be `native`. Here, we need to remove the first two characters from a string.

Note: `n` must be less than the length of the string.

4. Write a function to return `True` if the first and last number of a given list is same. If numbers are different then return `False`.

Expected Output:

```
Given list: [10, 20, 30, 40, 10]

result is True

numbers_y = [75, 65, 35, 75, 30]

result is False
```

5. Given two list of numbers, write a program to create a new list such that the new list should contain odd numbers from the first list and even numbers from the second list.

Given:

```
list1 = [10, 20, 25, 30, 35]
list2 = [40, 45, 60, 75, 90]
```

Expected Output:

```
result list: [25, 35, 40, 60, 90]
```

6. Write a function called exponent(base, exp) that returns an int value of base raises to the power of exp.

Note here `exp` is a non-negative integer, and the base is an integer.

Expected output

Case 1:

```
base = 2

exponent = 5
```

```
2 raises to the power of 5: 32 i.e. (2 *2 * 2 *2 *2 = 32)
```

Case 2:

```
base = 5

exponent = 4
```

```
5 raises to the power of 4 is: 625
```

```
i.e. (5 *5 * 5 *5 = 625)
```

7. Write a program to create function `calculation()` such that it can accept two variables and calculate addition and subtraction. Also, it must **return both addition and subtraction in a single return call**.
8. Given two strings, `s1` and `s2`. Write a program to create a new string `s3` by appending `s2` in the middle of `s1`.

Given:

```
s1 = "Ault"

s2 = "Kelly"
```

Expected Output:

```
AuKellylt
```

9. Create an inner function to calculate the addition in the following way
   - Create an outer function that will accept two parameters, `a` and `b`
   - Create an inner function inside an outer function that will calculate the addition of `a` and `b`
   - At last, an outer function will add 5 into addition and return it.
10. Write a program to create a function `show_employee()` using the following conditions.
    - It should accept the employee's name and salary and display both.
    - If the salary is missing in the function call then assign default value 9000 to salary

You need to done with your exercise within given time.